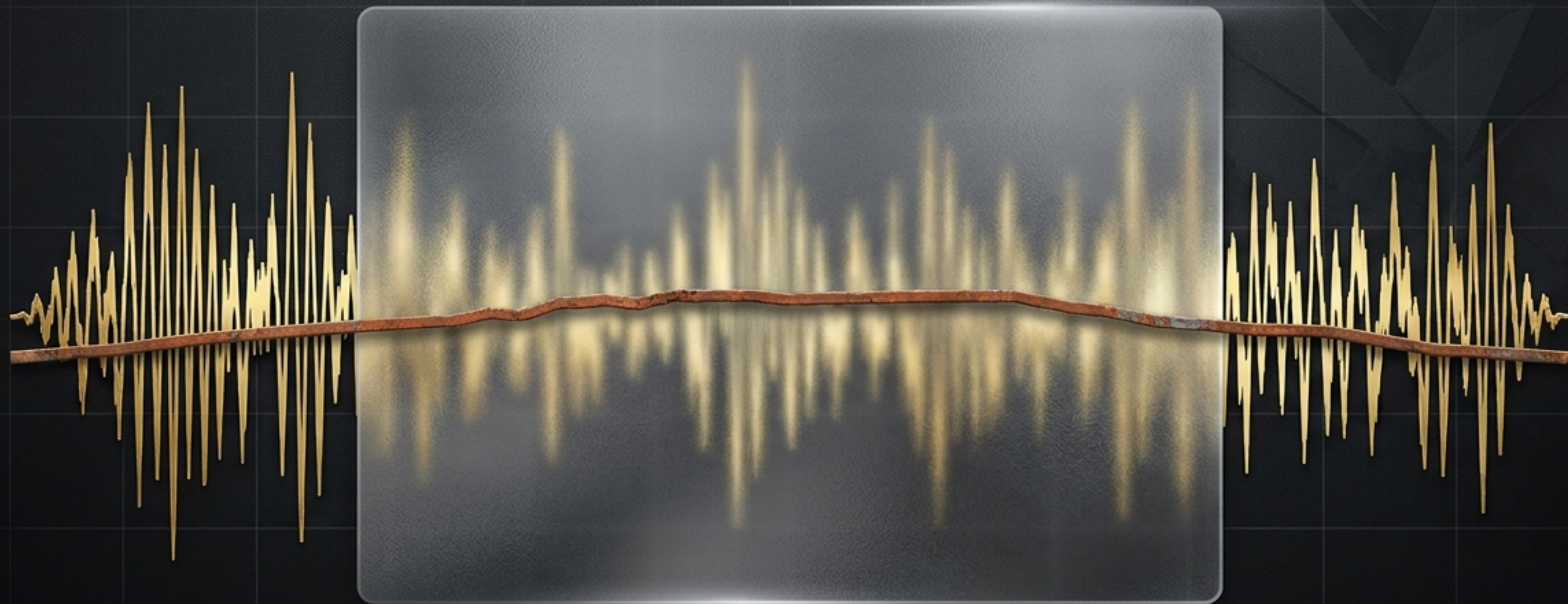




THE MASKING EFFECT

Exposing the structural failure of Fixed-Rank Kriging and the capital trap of spatial interpolation

MPDOK Mining Lab | USGS NURE-HSSR Stream-Sediment Geochemistry | Carlin Trend, Nevada

Invisible **Gold** and the Geospatial Reliance

The Arena



The Paradigm

Geochemical Data



Spatial Interpolation Model
(Fixed-Rank Kriging)

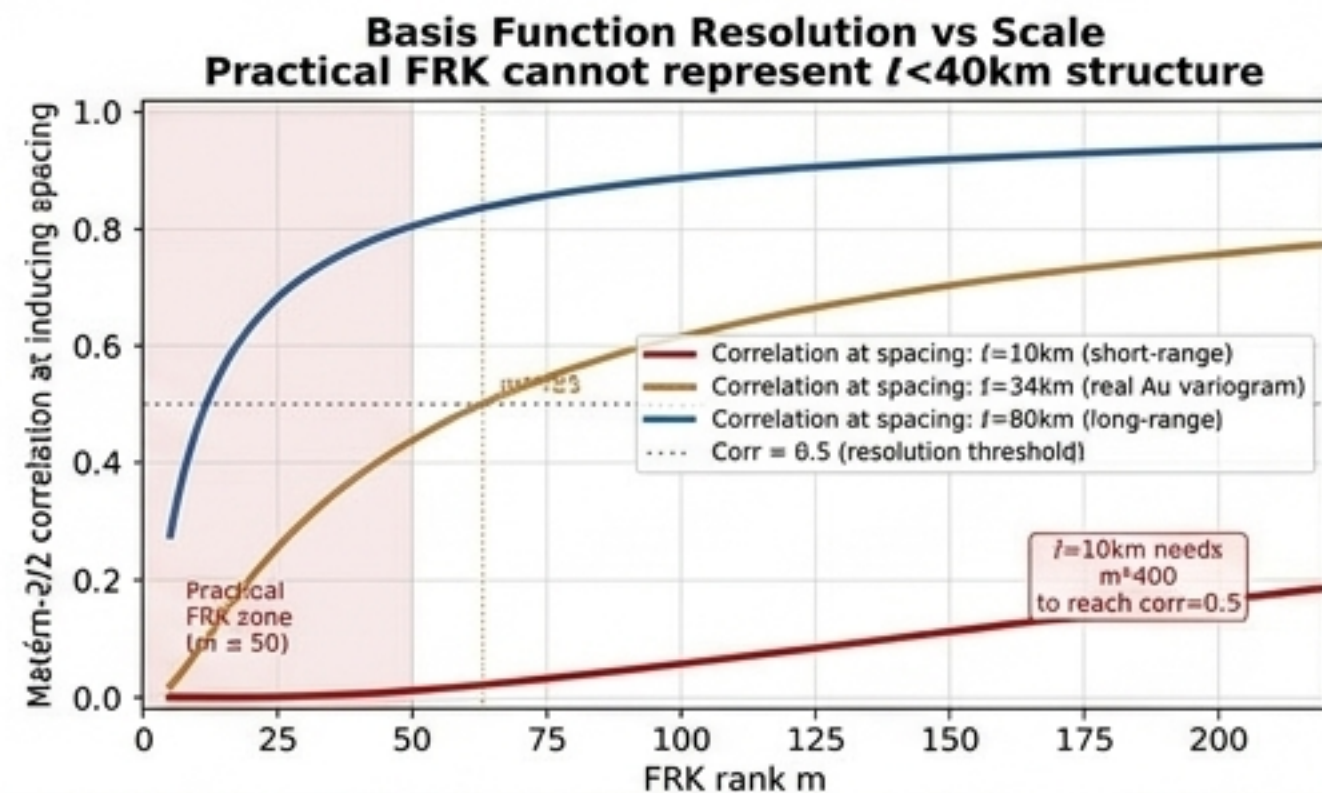
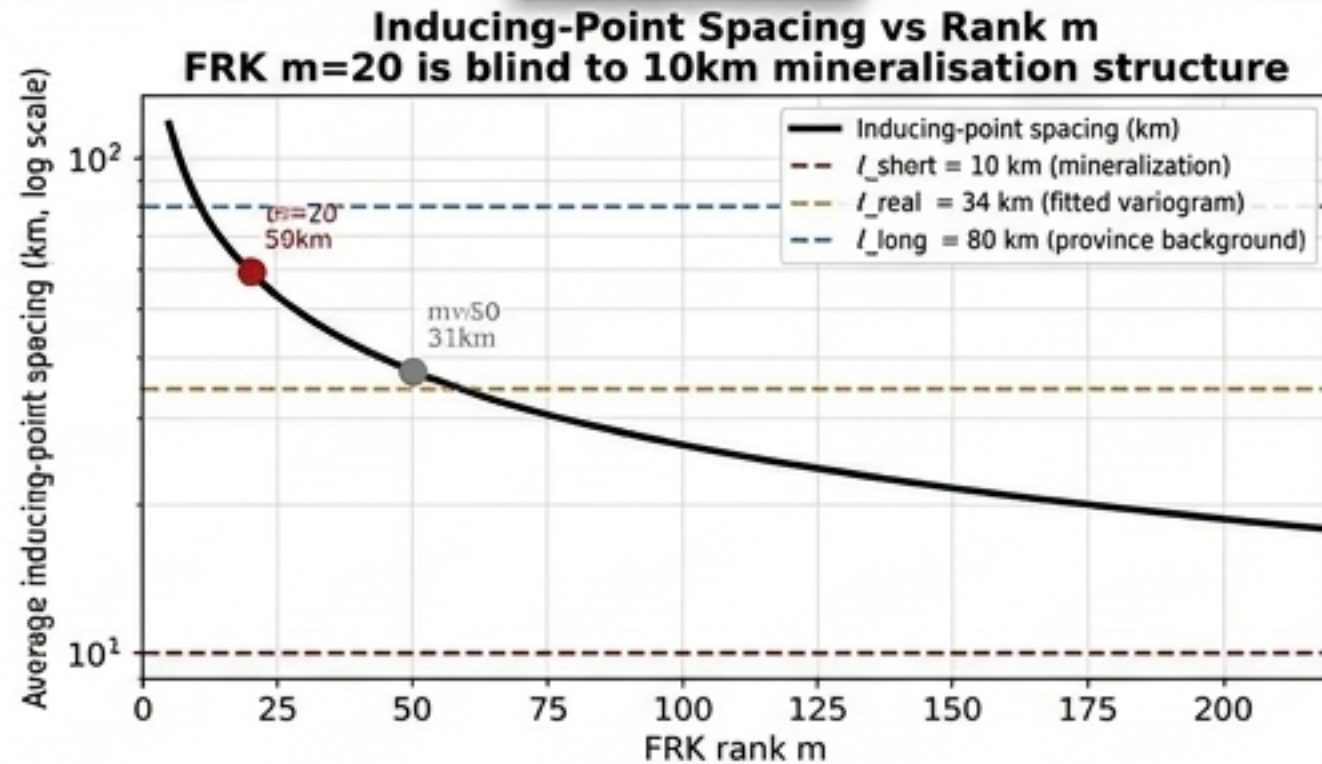


Multi-Million Dollar
Drilling & Milling Decisions

The **industry standard** for large spatial datasets is **Fixed-Rank Kriging (FRK)**—a low-rank approximation that uses m inducing points to predict grade between samples. But what if m points **cannot physically catch the gold**?

The Geometric Flaw of Fixed-Rank Kriging

The Math



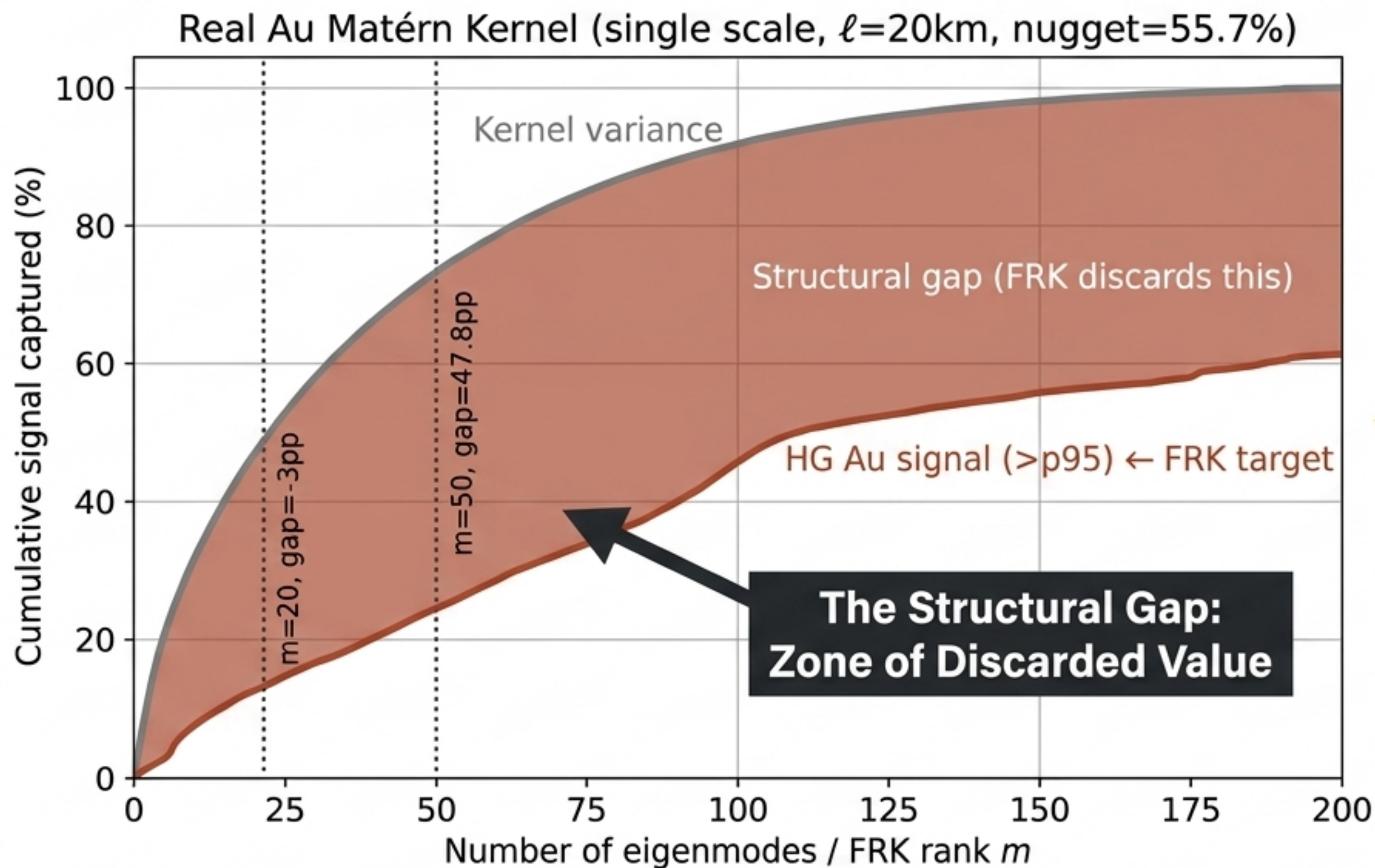
The Physical Metaphor

For a 69,000 km² domain, $m=20$ inducing points space out to **59.2 km**.



A **59 km net cannot catch a 10 km fish**. The Matérn correlation at this spacing is 0.0004—essentially zero. FRK basis functions are physically orthogonal to the high-grade signal.

The Spectral Proof: Discarding the Ore



The eigenspectrum
proves the **physical limit**.

The top **50** eigenmodes
capture **70.7%** of the total
variance, but only **22.9%**
of the high-grade signal.

The gold lives in the
spectral tail that FRK
permanently **deletes**.

The Paradox of the Masked Failure

If FRK cannot physically resolve high-grade ore, why does the industry still use it?

The 55.7% Nugget Effect

Answer Panel

44.3% Structural Signal
(Range = 34.4 km)

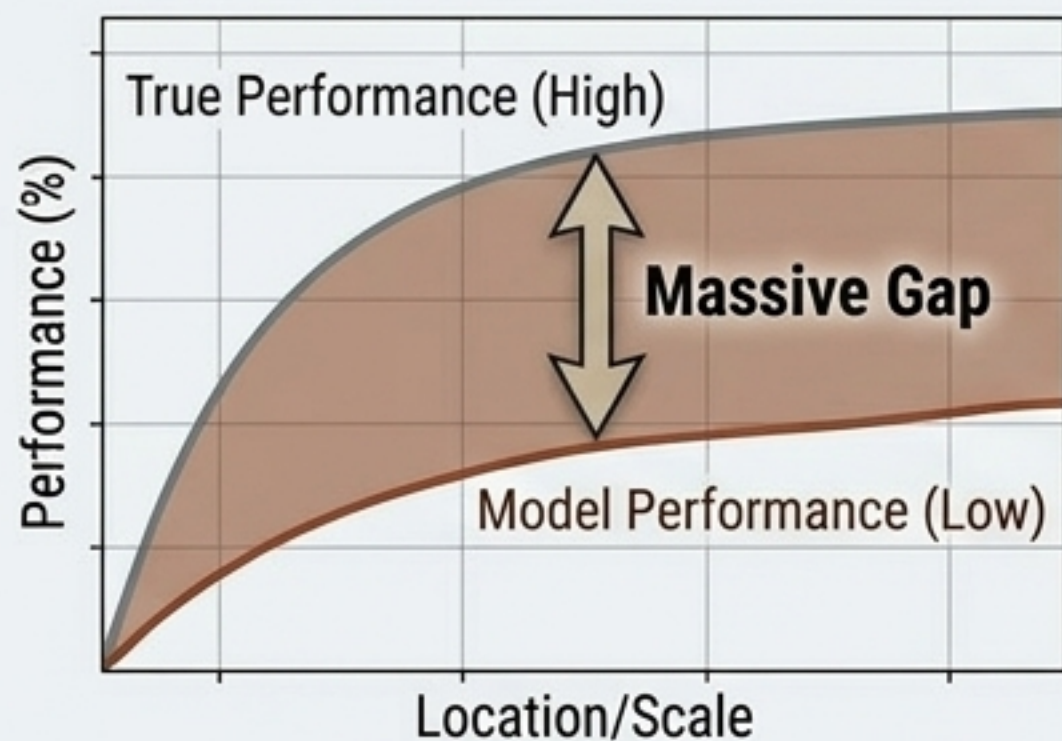
55.7% Nugget Effect
(Pure, uncorrelated white noise from
sampling error and micro-scale geology)

100% of Total Variance of the fitted Matérn-3/2 variogram

Both FRK and full-rank models hit this exact same noise ceiling. The nugget creates an illusion of parity.

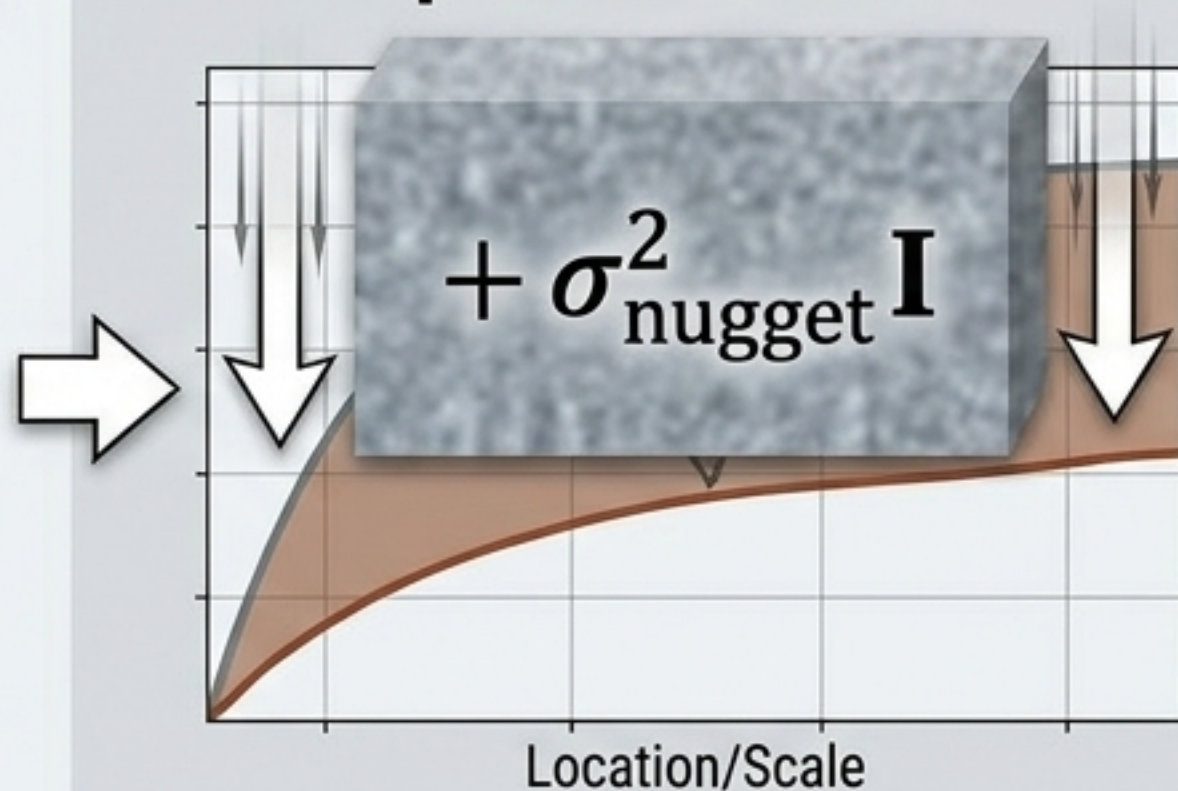
The Mechanism of the Mask

Step 1: The Baseline



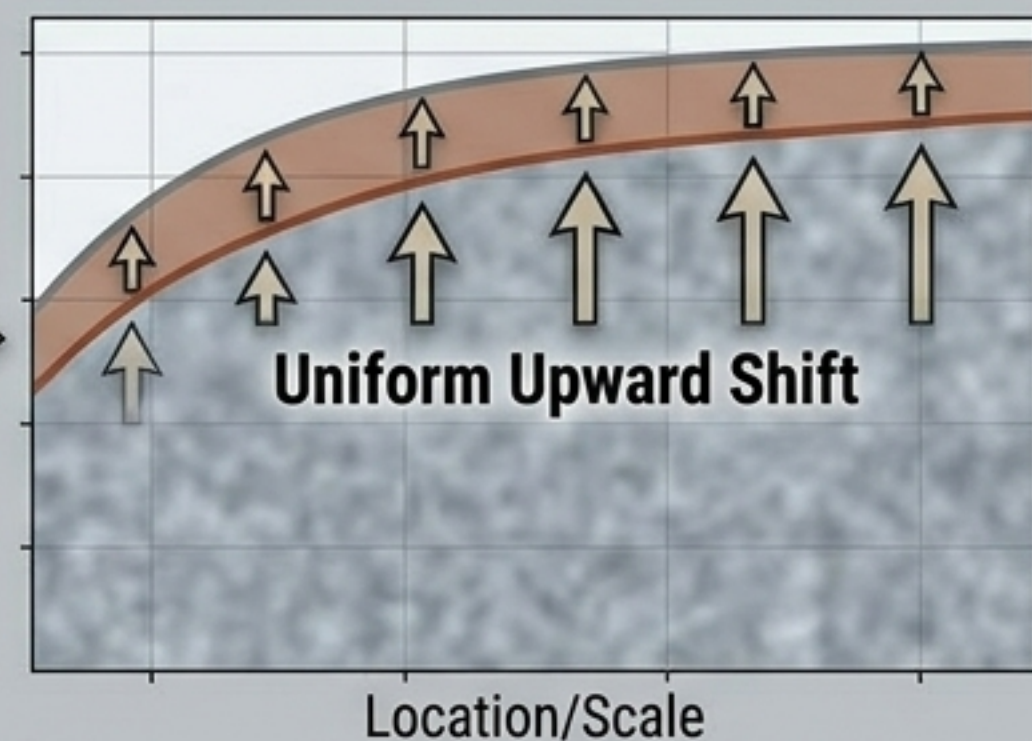
A massive performance gap exists.

Step 2: The Mask



$+\sigma^2_{\text{nugget}} \mathbf{I}$ is dropped onto the kernel.

Step 3: The Compression

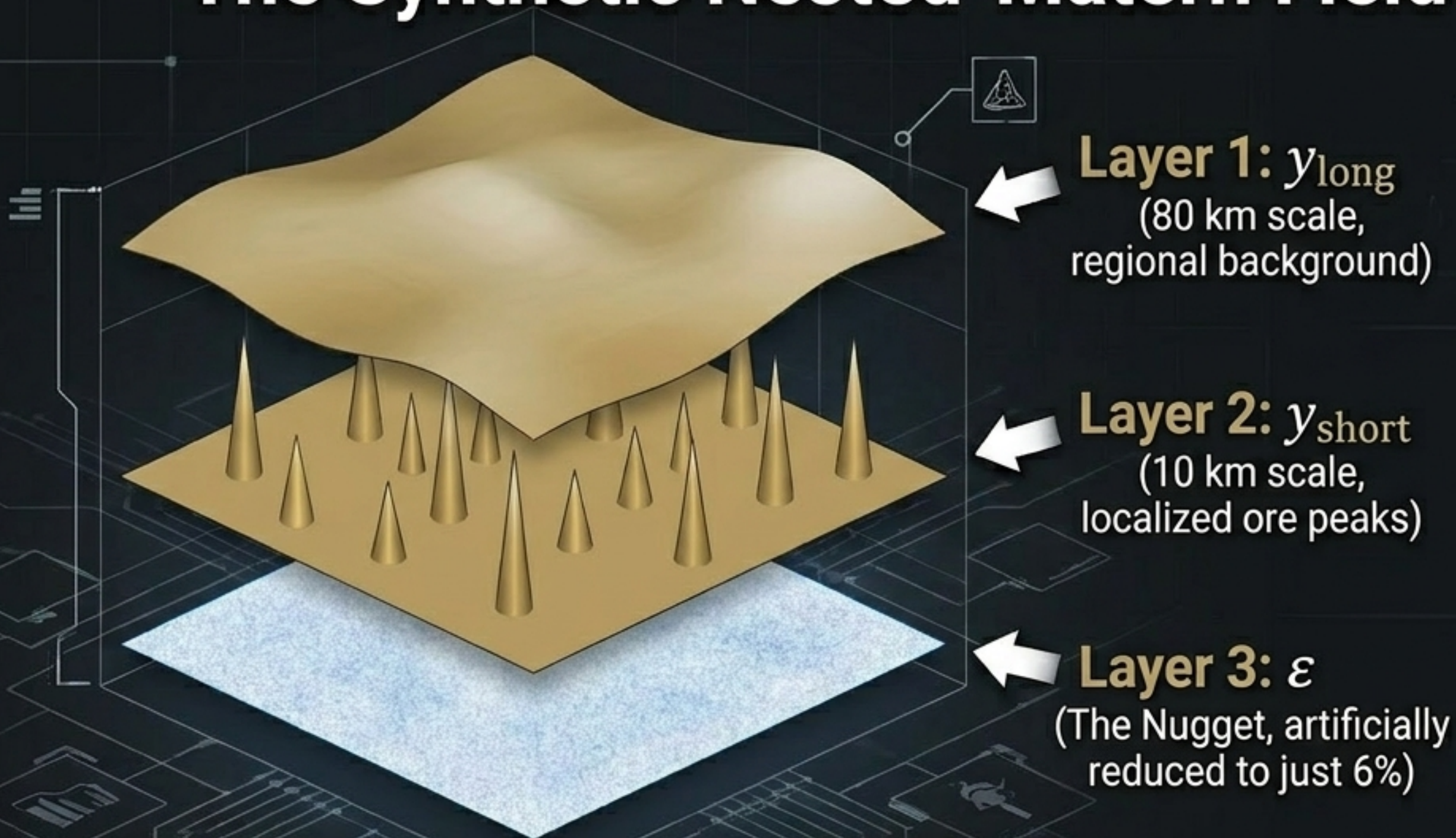


Every eigenvalue shifts upward uniformly, compressing relative differences.

In cross-validation on real data, this **55.7%** noise floor compresses the performance gap at high-grade locations to just **12.8 percentage points** (MPDOK **23.0%** vs FRK **10.2%**). The failure is hidden in the noise.

Removing the Noise Floor

The Synthetic Nested-Matérn Field Construction

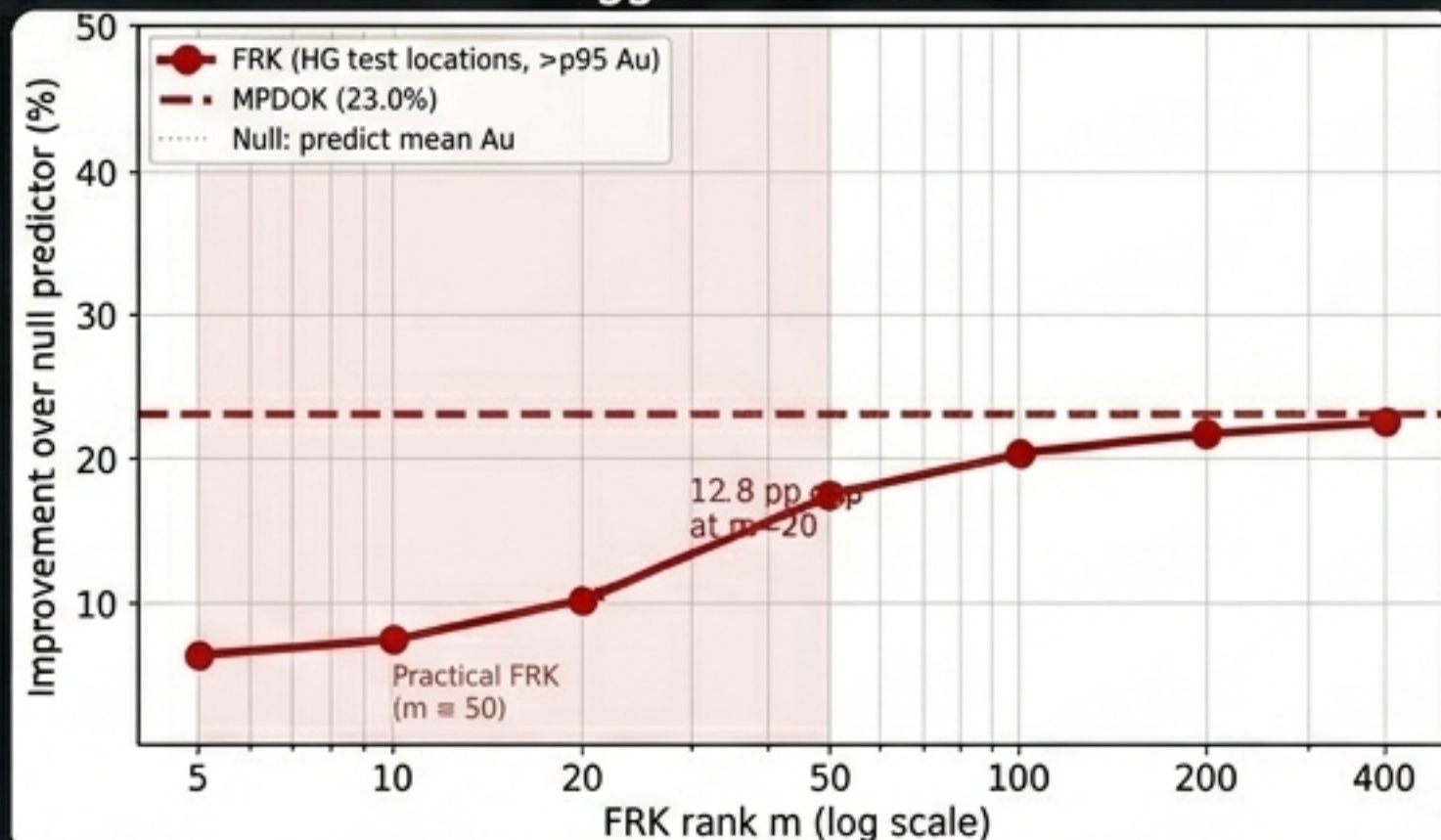


This synthetic field represents what high-quality, block-averaged drillhole data or filtered airborne geophysics would look like—spatial structure without the blinding white noise of raw stream sediments.

The Unmasking of the Structural Gap

The Illusion

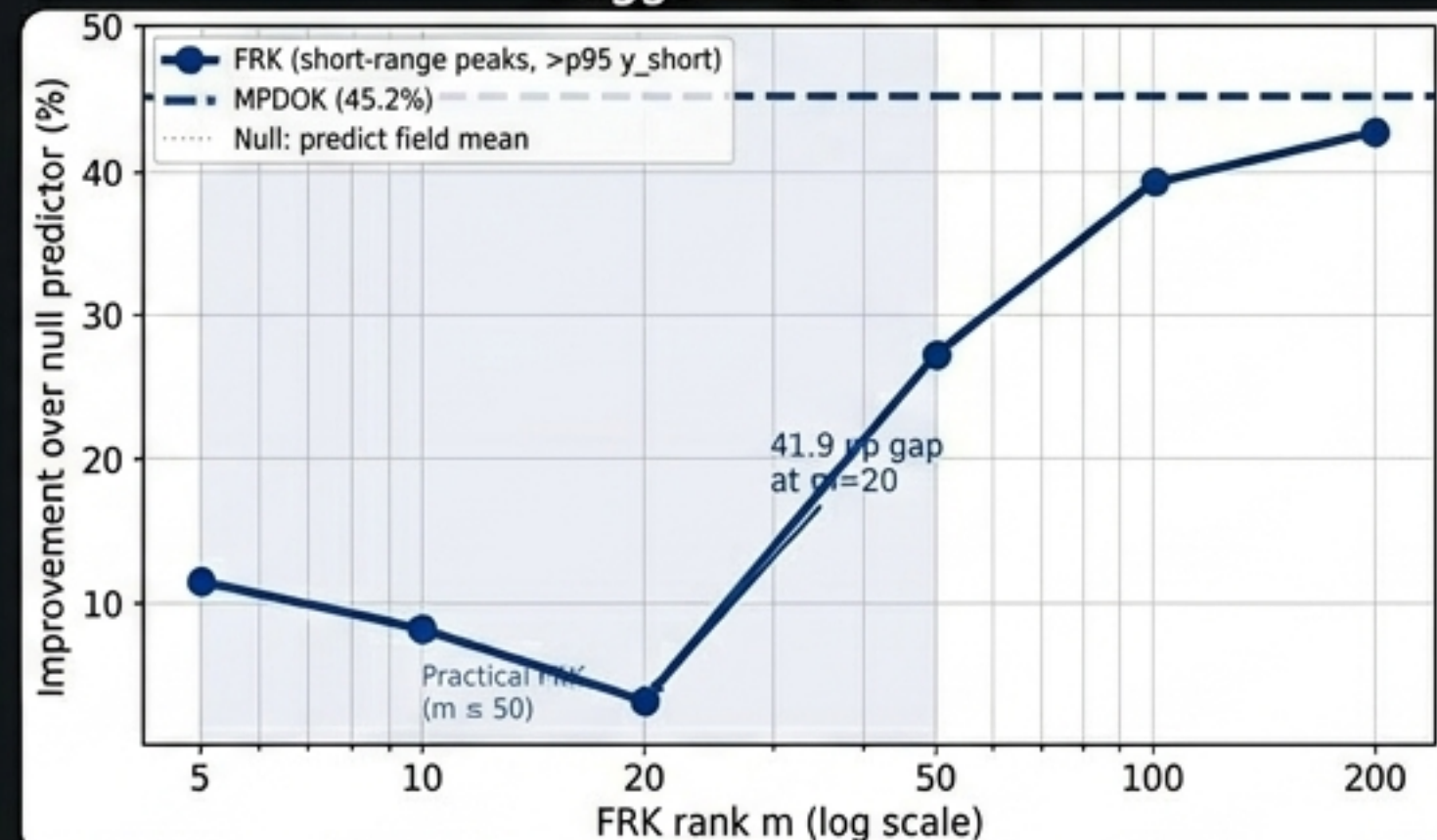
Real NURE-HSSR Au Data
Nugget = 55.7% of sill



Real NURE-HSSR Au Data (55.7% Nugget)
Gap at $m=20$ is a compressed 12.8 pp.

The Truth

Synthetic Nested-Matérn Field
Nugget = 6.0% of sill



Synthetic Nested-Matérn Field (6% Nugget)
The gap explodes to 41.9 pp.

The nugget was masking the failure. The failure was always there.

Blindness at the Peaks



FRK m=20

$$r = -0.05$$

Equivalent to random guessing.



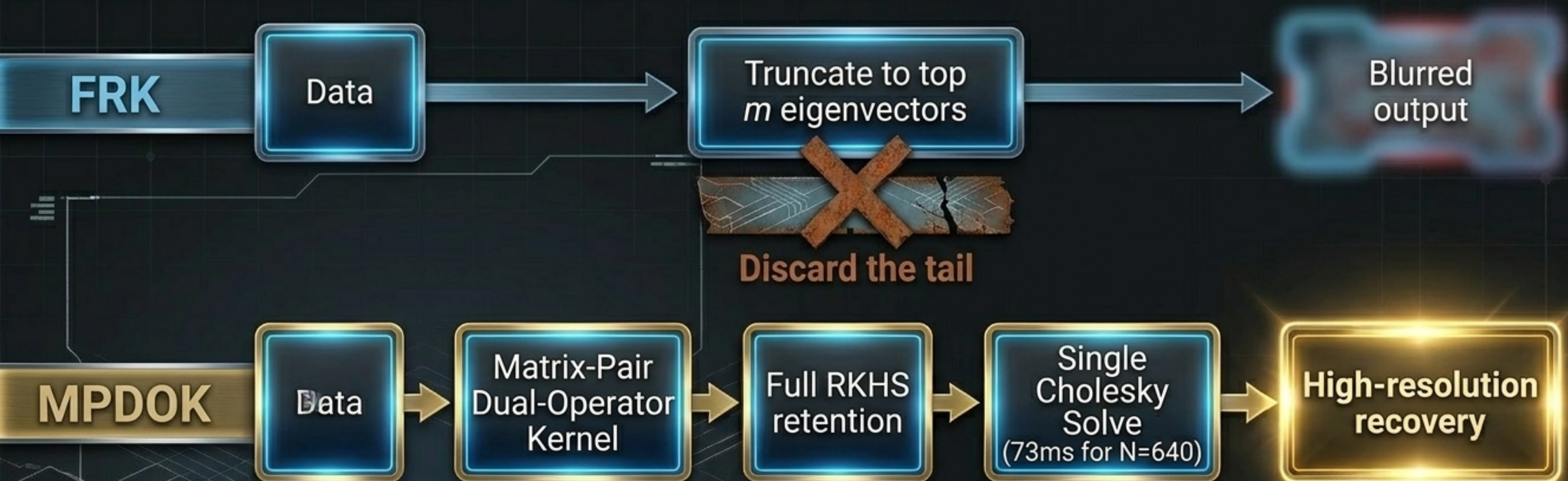
MPDOK

$$r = 0.92$$

Near-perfect high-frequency recovery.

At the exact 10km spatial scale where high-grade economic anomalies live, operational FRK is entirely blind.

The Full-Rank Recovery of MPDOK



MPDOK solves the $O(N^3)$ bottleneck entirely without a rank parameter. It retains both the smooth geological background and the localized anomalies. No truncation. No lost value.

The Capital Trap Part I: The Drilling Explosion

Relying on FRK for
10km Ore Targets

Expand $m > 400$ to achieve
physical resolution threshold.

To mathematically support $m > 400$ without matrix ill-conditioning requires an exponential explosion in spatial sampling.

Financial Impact: Unnecessary infill drilling campaigns costing \$50–\$300 per metre. A massive capital expenditure simply to compensate for a mathematical algorithm's architectural flaw.

The Capital Trap Part II: The Cost of the Blur

Accept operational
FRK at $m=20$

The model acts as a low-pass
spatial filter, smearing high-grade
peaks into surrounding rock.

Revenue Loss

High-grade ore is diluted below the cut-off grade. The model classifies it as sub-economic waste.

Gold is left in the ground.

Milling Waste

Smooth predicted-grade surfaces extend moderate values into barren areas. The team blasts and mills worthless rock at an operational cost of \$10–\$30 per tonne for zero yield.

Diagnostic Matrix: FRK vs. MPDOK

Dimension	Fixed-Rank Kriging (m=20)	MPDOK
Architecture	Truncated low-rank approximation	Full RKHS projection (no truncation)
Geometric Capability	Blind to 10km mineralization (spacing = 59km) ⚡	Resolves all scales natively ✓
Peak Correlation	$r = -0.05$ (Random)	$r = 0.92$ (Highly accurate)
Financial Outcome	Forces excessive drilling or ore/waste misclassification	Maximizes capital efficiency & ore recovery

A Universal Failure Pattern

Using a rank-k proxy to avoid $O(N^3)$ covariance costs systematically deletes the high-frequency events that dictate outcomes.

Aerospace
(SQ321)

EnKF k=50 ensemble
+ Gaspari-Cohn

Erases localized
wind-shear

Fatal turbulence.

Mining
(Carlin Trend)

FRK m=20
inducing points

Erases
short-range ore

Capital loss.

Genomics
(APY)

m core animals
approximating GRM

Erases rare
variants

BLUP accuracy
ceiling.

Finance

Hop-limited
correlation graph

Erases indirect
correlations

Unhedged tail
exposure.

MPDOK universally resolves the high-frequency signal without rank truncation.

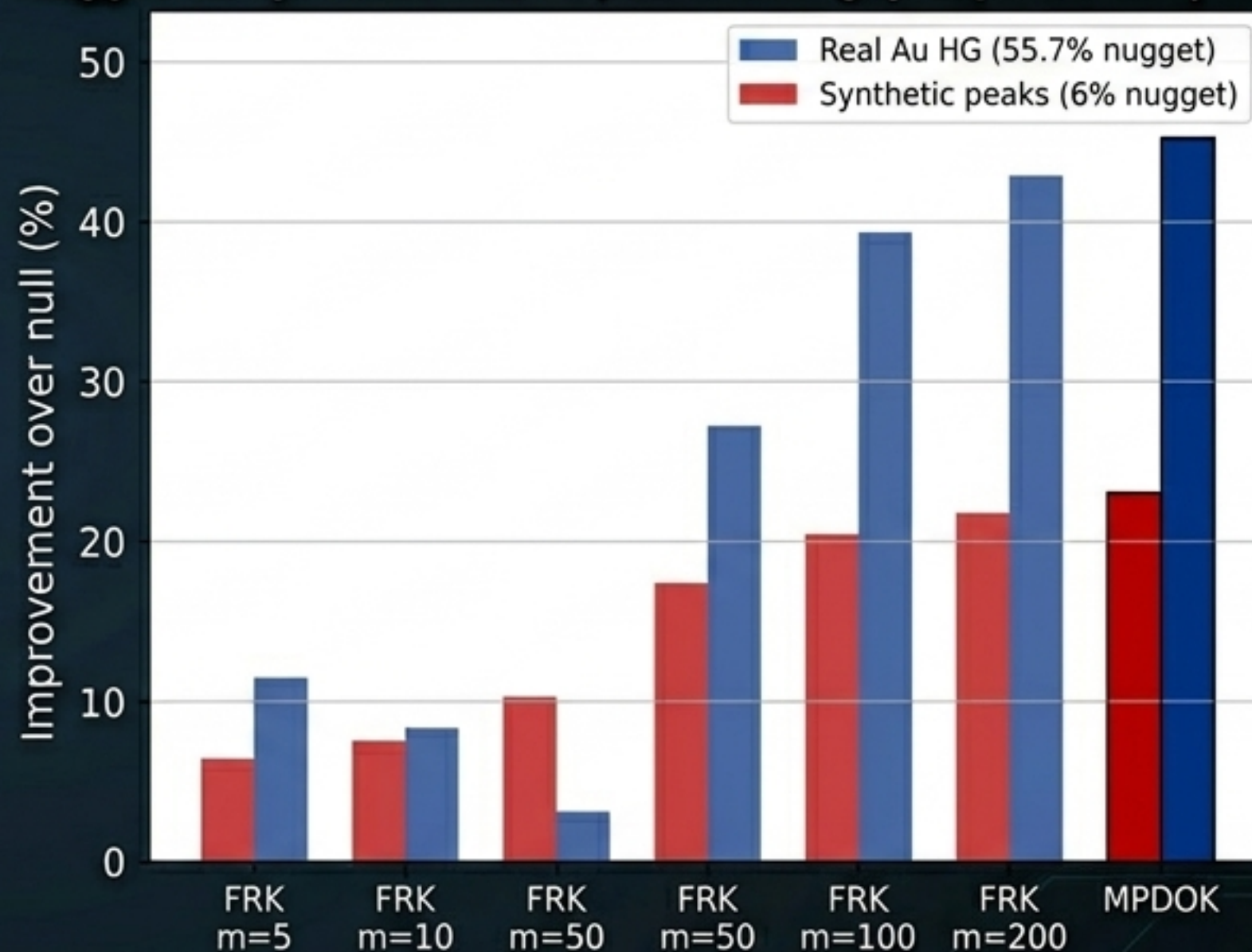
The Failure Was Always There. MPDOK Exposes It.

The Geometric Fact: FRK basis functions are physically incapable of resolving localized high-grade ore. This is a constraint of geometry, not sample size.

The Masking Illusion: Massive nugget effects in raw geochemical data simply compress relative errors, hiding the structural flaw.

The Capital Imperative: MPDOK operates in the full RKHS in milliseconds, eliminating the FRK-induced capital trap of excessive drilling and milled waste.

Summary: all methods, both datasets.
Nugget compresses real Au; structural gap exposed in synthetic.



Stop interpolating the noise. Start recovering the ore.